3.0 SPACETIME and MOTION

-ŠŘ- 'SPACETIME / SPACE / PASSAGE OF TIME'		Associated Affix: **STS**
	STEM 1	**STEM 2:** [the spatial component of Stem 1, i.e.,] (to be/occupy) a location in space, the site of something; to be spatially located/situated at, to be the site of something **STEM 3:** [the temporal component of Stem 1, i.e.,] (to occur) at a particular point/duration/period in time; to temporally take place at
BSC	(to exist/occur at) a "location/period/instance" of spacetime; to exist/occur at a place and time [both the spatio-temporal location and the entity occupying it]	
CTE	one's physical and temporal existence; to spatio-temporally exist [i.e., exist at a particular place at a particular time]	
CSV	(to be) a particular volume/duration of spacetime; (to be) a particular location/volume of space at a particular moment/period in time	
OBJ	(to be) an entity located at a particular place and time; (to be) a spatio-temporally existent entity	

-LC- 'PATHWAY / COURSE / ROUTE / ROADWAY / CHANNEL'		
BSC	**Stem 1:** (to be) a marked/indicated pathway/trail/course	**Stem 2:** (to be) a marked route / roadway / street **Stem 3:** (to be) a marked/indicated channel [i.e., planned conduit-like courseway along/through which an entity/substance travels]
CTE	(to be) the direction/destination to which a pathway/trail leads (i.e., what it's there for)	
CSV	(to be) the physical manifestation of a pathway/trail (i.e., how it is physically marked/distinguished from the surrounding surface or underlying substrate)	
OBJ	(to be) a party/entity which travels a pathway/trail/course	

-KÇN- 'MAP / DIAGRAM / BLUEPRINT'		
BSC	**Stem 1:** (to be) an act/instance of reading/using a map or map-like chart; to read/use a map or map-like chart [i.e., a miniaturized graphic representation of a localized geographic area or landscape or spatial volume or ocean volume, etc.]	**Stem 2:** (to be) an act/instance of reading/using a diagram, flow-chart, or other graphic representation of the steps/relationships/interdependencies of a sequential process or process flow; to read/use a diagram **Stem 3:** (to be) an act/instance of using/reading/interpreting a blueprint; to use/read/interpret a blue print [i.e., a formalized/idealized, abstract visual 2-dimensional or 3-dimensional model of an object/entity for purposes of understanding its structure/construction/architecture
CTE	(to be) the physical map itself	
CSV	(to be) a physical act of reading/using/interpreting a map; to read/use/interpret a map	
OBJ	(to be) what a map represents, the geographic/spatial area represented on/by a map	

3.1 SPATIO-TEMPORAL DIMENSIONS

-**JG**- **'DIMENSIONAL ATTRIBUTE'** (use with the SUF, EXD, FLC, PTW, ITY, etc. affixes)		
	STEM 1	**STEM2:** (to be) a decrease in the degree of a particular dimensional attribute **STEM 3:** (to be) an increase in the degree of a particular dimensional attribute
BSC	(to be) a degree of a (static) dimensional attribute (e.g., length, width, volume, etc.)	
CTE	(to be) the intrinsic nature/essence of a particular attribute (i.e., the existence of the attribute within or as part of an entity regardless of its physical manifestation	
CSV	(to be) the physical manifestation of a particular dimensional attribute	
OBJ	(to be) the object/entity having the particular dimensional attribute	

The Stems and Specifications of the following roots are modeled on the -**JG**- 'DIMENSIONAL ATTRIBUTE' root above:

-**ẒB**- 'PLANARITY/FLATNESS RELATIVE TO THE FORM OF AN OBJECT ITSELF' [i.e., the extent to which all the points of a surface are parallel to each other in the dimensional context of the surface itself, such as the surface of a blanket (regardless of whether it is folded or laid flat) or the surface of a sphere (where the sphere's surface can be considered flat/planar from the perspective of the surface itself, regardless of the fact the sphere as a whole is not]

-**ČF**- 'WIDTH/SLENDERNESS' [i.e., the linear distance between the two points furthest from each other on a cross-sectional plane through an object]

-**ẒD**- 'GIRTH/CIRCUMFERENCE'

-**JN**- 'LENGTH' (i.e., the longest spatial dimension of an object/entity)

-**CH**- 'HEIGHT (= TALLNESS PARALLEL TO THE DIRECTION OF GRAVITY)

-**VJ**- 'AMPLITUDE'

-**ZẒ**- 'BINARY POLARITY' (i.e., the degree to which something reverses direction of a binary attribute, e.g., rotation, spin, charge, etc.)

-**KČ**- 'REACH' (i.e., the degree to which something extends outward in a quasi-linear manner)

-**ẒX**- 'SPREAD' (i.e., the degree to which something extends outward in 2-dimensional quasi-planar coverage)

-**VČ**- 'EXTENT' (i.e., the degree to which something extends outward as a three-dimensional volume)

-**BẒ**- 'DEPTH' (i.e., the shortest spatial dimension of an object/entity)

-**ŠL**- 'THINNESS/THICKNESS (i.e., ratio of shortest dimensional end-to-end axis-length to next longest dimensional end-to-end axis-length)'

-**JḶ**- 'SPATIAL GAP'

-**GẒ**- 'TEMPORAL INTERVAL' [i.e., the elapsed time/duration between two events/occurrences/states]

-**FC**- '3-DIMENSIONAL VOLUME'

-**ŢC**- ‘RECTILINEAR/POLYHEDRAL/STRAIGHT-ANGLED’

-**GJ**- ‘ROUNDEDNESS’ [i.e., the extent to which a surface is spheroidally curved from an external perspective]

-**CF**- ‘FLATNESS/LEVELNESS OF A SURFACE’ [i.e., the extent to which the points of a surface constitute a flat plane from an external perspective]

-**CŢ**- ‘2-DIMENSIONAL AMORPHOUS/SHAPELESS’

-**Ẓ Ţ**- ‘3-DIMENSIONAL AMORPHOUS/SHAPELESS’

-**ẒḌ**- ‘LINEAR (i.e., ONE-DIMENSIONAL) CURVATURE’

-FKL- ‘PHYSICAL SPATIO-TEMPORAL EXTENSION’		
BSC	**Stem 1:** (to be) a state/act of spatio-temporal extension linearly/uni-dimensionally (up/out/along/back, etc.) to a certain point/level/height, etc.; to extend in such a manner, to“reach” (to) a certain point/level/height, etc.	**Stem 2:** (to be) a state of extension two-dimensionally (out/among/away, etc.) to a certain distance/edge/linear landmark/linear boundary, etc., “reach” (to) a certain extent/distance/edge/linear landmark/boundary, etc.; to extend in such a manner **Stem 3:** (to be) a state of spatio-temporal extension three-dimensionally (out/among/away, etc.) to a certain distance/planar edge/planar landmark/planar boundary, etc., “reach” (to) a certain distance/planar edge/planar landmark/planar boundary, etc.; to extend/ “reach” three-dimensionally in such a manner
CTE	(to be) a state of extension / being extended unidimensionally	
CSV	(to be) an act of spatio-temporal extension unidimensionally	
OBJ	(to be) that which spatio-temporally extends or “reaches” unidimensionally to a particular point/level/height, etc.	

-TŇ- ‘LEVEL / FLOOR / TIER’		
BSC	**Stem 1:** (to be in) a position at/on a particular planar level of a vertically sequential series of levels; to be positioned on a particular level (of a vertically sequential series of levels)	**Stem 2:** (to be in) a position on a particular floor/story of a building (e.g., be on the fifth floor/story of a hotel); be on a particular floor/story of a building **Stem 3:** (to be in) a position on a particular abstract level/tier of a series of metaphorical/abstract tiers/levels of something (e.g., ‘He plays at grandmaster level/tier when it comes to chess.’)
CTE	(to be) in a state of being on a particular physical level of a vertically sequential series of levels	
CSV	(to be) an act of physically positioning oneself or being positioned on a particular level	
OBJ	(to be) the particular physical level something or someone is positioned on	

-TŇW- ‘ROW / FILE / QUEUE’		
BSC	**Stem 1:** (to be in) a position in a particular linear/planar row of a horizontally sequential series of rows; to be positioned in a particular row (of a horizontally sequential series of rows) [as viewed externally from the row]	**Stem 2:** (to be in) a position in a particular linear/planar rank/file/line of a sequential series of front-to-back ranks/files/line; to be positioned in a particular rank/file (of a sequential series of front-to-back ranks/files) [as viewed externally] **Stem 3:** (to be in) a position in a queue; to be (lined up) in a queue
CTE	(to be) in a state of being in a particular physical row of a horizontally sequential series of row	
CSV	(to be) an act of physically positioning oneself or being positioned on a particular row	
OBJ	(to be) the particular physical row something or someone is placed/positioned in	

-ŘTŇ- ‘COLUMNAR POSITION’		
BSC	**Stem 1:** (to be in) a position in a particular linear vertical column of a sequential series of horizontally-aligned columns; to be positioned oin a particular vertical column (of a sequential series of horizontally-aligned columns)	**Stem 2:** (to be in) a position in a particular linear vertical column of a sequential series of front-to-back-aligned columns; to be positioned oin a particular vertical column (of a sequential series of front-to-back-aligned columns) **Stem 3:** (to be in) a position in a vertical column (i.e., a vertical/columnar queue)
CTE	(to be) in a state of being in a particular physical vertical column of a sequential series of horizontally-aligned columns	
CSV	(to be) an act of physically positioning oneself or being positioned in a particular physical vertical column of a sequential series of horizontally-aligned columns	
OBJ	(to be) the particular physical column something or someone is positioned in	

-ŇŽ- ‘ARRANGEMENT / ORDER / SYMMETRY’ Affix: **CLS**		
BSC	**Stem 1:** (to be) a state/act/instance of arranging or being arranged (i.e., place(d) in some sort of order or discernible pattern); to arrange, to order	**Stem 2:** (to be) a an act/state/instance of arranging/ordering or being arranged/ordered based on a specific criterion or criteria **Stem 3:** (to be) an act/instance of symmetrical arrangement; to arrange symmetrically
CTE	(to be) a particular arrangement or pattern	
CSV	(to be) an act/process of physically placing entities into an arrangement or pattern	
OBJ	(to be) an entity in or part of a particular arrangement or pattern	

MORPHOLOGICAL DERIVATIVES: array, setting, hierarchy, symmetry, disarray, scatter, mess, chaos

-ŇŽW- 'REGULARITY / CLASSIFICATION / ASSORTMENT'		
BSC	**STEM 1:** (to be) a state/act/instance of being/making regular or displaying regularity (i.e., having an expected/conventionalized/predictable pattern); be/make regular, regularize	**STEM 2:** (to be) a state/act/instance of being in or making a classification (i.e., an arranged scheme for storage-and-retrieval); to classify
CTE	(to be) a state of being regular	
CSV	(to be) an act/process of making something regular; to regularize	**STEM 3:** (to be) a state/act/instance of being in or making an assortment (i.e., an arranged selection of members/components); to sort
OBJ	(to be) something that is in a state of being regular or having been regularized	

MORPHOLOGICAL DERIVATIVES: regularity, regulate

3.2 MOTION AND SPATIAL POSITION

English can create succinct "moving images" of complex trajectories, paths, treks, and in-situ movements by simple juxtaposition of locative/directional particles (i.e., prepositions acting mostly as adverbial particles). For example , imagine the manager of a basement nightclub talking to a friend on the sidewalk outside his club. He invites the friend to the small patio outside the back of the club, which one finds by descending into the club, going past the dance floor, up a flight of stairs, and down a long hallway to the back door. He does so by saying,'*Come on down inside up along through out back to the patio.'*

New Ithkuill allows one to create similar succinct "moving images". Each motion root (whether referring to in-situ or translative motion) has a corresponding -$\mathbf{V_XC_S}$ affix of the same form (e.g., the root -**KR**-for 'OBLIQUE LINEAR MOTION' has a corresponding suffix -**kr**.) Each of these suffixes has the following nine degrees:

1	Interior perlative motion/movement in a manner described by the suffix's corresponding root among/within an external/surrounding/underlying (quasi-)static medium. For translative motion suffixes, implies motion along a course described by the suffix's corresponding root where the course is set, determined, or prescribed by an externally surrounding/underlying medium/channel, etc.
2	Interior motion/movement: static or translative movement/motion in a manner described by the suffix's corresponding root while inside another object/entity (where, unlike the perlative degree, the exterior object/entity does not define/prescribe/determine the course of the moving object/entity
3	Implies entry from an exterior space/medium/context into an interior space/medium/context
4	Implies continued/continuous movement in a manner described by the suffix's corresponding root; for translative motion suffixes, implies continued/continuous motion along a course/direction described by the suffix's corresponding root
5	Implies a single motion in a manner described by the suffix's corresponding root across from one point/side of the contextual space to another; for translative motion suffixes, implies a single motion along a course/direction described by the suffix's corresponding root across a gap/gulf/space from one place to another
6	For translative motion affixes, signifies motivational "on" as in "come on", "go on", "move on" - implies/reinforces need/impetus to begin/keep moving in the manner or direction described by the suffix's corresponding root without stopping. For static positional affixes, operates merely as a locative to specify an entity's X/Y/Z position.
7	Implies exit from an interior space/medium/context out to an exterior space/medium/context
8	exterior motion/movement: static or translative movement/motion in a manner described by the suffix's corresponding root while outside another object/entity (where, unlike the perlative degree, the other object/entity does not define/prescribe/determine the course of the moving object/entity
9	exterior perlative motion/movement outside of, over, next to, alongside, "above"/"under" an adjacent (quasi-)static surface/medium; for translative motion suffixes, implies motion on a course described by the suffix's corresponding root along, over, parallel to, around, "above"/"under", or tracking/following an adjacent (quasi-)static surface/medium

For Degree 6, the various Modulative suffixes as well as Phase can be used to describe repeated, ongoing movement/motion, e.g., back-and-forth, to-and-fro, side-to-side, up-and-down, in-and-out, out-and-back, etc.

Besides movement/motion roots, roots denoting any spatial referent/area would also have corresponding affix-forms utilizing this pattern (e.g., front side, rear area, top area/space, area below/underneath, etc.).

3.3 TRANSLATIVE MOTION ROOTS

-**TR**- **'PATH-ORIENTED, QUASI-LINEAR TRANSLATIVE MOTION'** Affix: **M01**		
	Stem 1	**Stem 2:** As per Stem 1 w/ perspective is on motion to(ward) the topical referent of the sentence or clause, i.e., "come; approach"; CPT = "arrive" **Stem 3:** As per Stem 1 w/ perspective is on motion away from the topical referent of the sentence or clause, i.e., "go; go away; move away"; CPT = "leave; depart"
BSC	act of translative motion from one spacetime location to another; to move from one place to another [both the motion and the entity moving]	
CTE	The process of movement itself from one place to another; to be moving from one place to another [focus on the movement, not the entity moving]	
CSV	The path/trajectory/course/route followed between two places; to move along/through the path/trajectory/course/route from one place to another	
OBJ	The entity moving from one place to another; to be the entity moving from one place to another	

Roots (and suffixes) following the same model as -**TR**- above include the following:

-**PR**- 'VERTICAL QUASI-LINEAR TRANSLATIVE MOTION', i.e., movement up/down; Stem 2 = 'ascent/ascend; going (straight) up; rise/raise/elevate'; Stem 3 = 'descent/descend; going (straight) down; "drop"; lower(ing)'

-**KR**- 'OBLIQUE VERTICAL QUASI-LINEAR MOTION', i.e., ascending/descending at an angle/slant'; Stem 2 = oblique ascent, upward incline, uphill; Stem 3 = oblique descent, downward incline, downhill

-**DR**- 'HORIZONTAL PLANAR TRANSLATIVE MOTION', i.e., movement along a line/path on a horizontal plane between the topical referent and a second location; Stem 2 = 'approach, getting nearer'; Stem 3 = 'recede/receding, getting farther away'

-**FR**- 'PARALLEL TRANSLATIVE MOTION', i.e., movement along same path/trajectory as the topical referent; Stem 1 = parallel movement in same direction alongside topical referent; Stem 2 = parallel movement alongside but in opposite direction to topical referent

-**BR**- 'PERPENDICULAR-PLANAR TRANSLATIVE MOTION', i.e., movement along a line/path "across" one's forward path or field of vision, moving laterally (i.e., horizontally perpendicular) to the positional/orientational position or path of the topical referent (i.e., analogous to watching something moving in a line across a screen held in front of one's eyes); Stem 2 = movement from left-to-right relative to the topical referent; Stem 3 = movement from right-to-left relative to the topical referent.

-**GR**- 'OBLIQUE LATERAL TRANSLATIVE MOTION', i.e., moving laterally at an oblique angle on same horizontal plane as the topical referent; Stem 2 = lateral movement at an oblique angle approaching (i.e., getting nearer to) the topical referent; Stem 3 = lateral movement at an oblique angle moving away from (getting farther away from) the topical referent.

-**GL**- 'HORIZONTAL-PLANAR RANGE OF MOTION', i.e., random directed movement within/throughout a 2-dimensional horizontal plane.

-**DL**- 'VERTICAL-PLANAR RANGE OF MOTION', i.e., random directed movement within/throughout a 2-dimensional vertical plane.

-**KL**- 'PERPENDICULAR-PLANAR RANGE OF MOTION', i.e., random directed movement within/throughout a 2-dimensional vertical plane perpendicular to the perspective of the topical referent (i.e., lying "across" one's visual field or directional path, analogous to a painting or flat screen held up in front of a person).

-**PL**- 'TRANSLATIVE MOTION ALONG PARABOLIC/ARC-LIKE TRAJECTORY RELATIVE TO GRAVITY'

-**BL**- 'MOTION IN A CURVE'

-**FL**- 'MOTION IN A PLANAR CIRCULAR PATH' [Stem 2 refers to counter-clockwise motion and Stem 3 to clockwise motion, relative to the speaker/observer]

-**TL**- 'GENERIC/OBLIQUE 3-DIMENSIONAL RANGE OF MOTION', i.e., 3-D pervasive movement throughout or within a volume of space

NOTE: When using the affix the above fourteen roots, to specify the Stem 2 meaning of the respective root, add -**w** to the affix form, e.g., -**tr** becomes -**trw**, -**kl** becomes -**klw**. To specify the Stem 3 meaning of the respective root, add -**y** to the affix form, e.g., -**tr** becomes -**try**, -**kl** becomes -**kly**.

3.4 SPATIAL POSITION/LOCATION ROOTS

-Ṭ- **'SPATIAL POSITION, LOCATION, ORIENTATION, DIRECTION'** Affix: **P01**		**STEM 2:** Direction relative to the observer (i.e., linear directional path between an object and the observer) **STEM 3:** the opposing direction or direction from which something emerges/originates/appears, etc.
BSC	**STEM 1:** spatial position of an entity; to be situated/located at	
CTE	The act of physically occupying a/the position/location in space; to physically occupy a/the position in space	
CSV	A/the position/location in space occupied by an entity; to be a/the position/location in space	
OBJ	The entity situated/located at; to be that which is situated/located at	

Stem Zero of this root = Area, general vicinity of a particular position/location relative to the observer, whether 2-dimensional or 3-dimensional according to the situational context

Roots following the same model as -Ṭ- above include the following based on a 3-dimensional X/Y/Z axial grid:

-**Ḍ**- 'POSITION/LOCATION AT 0 / 0 / 0', i.e., "right here"; at the center point of the observer's 3-dimensional spatial frame of reference.

-**ṬY**- 'POSITION/LOCATION AT 0 / 0 / +Z', 'above the observer'

-**ṬW**- 'POSITION/LOCATION AT 0 / 0 / -Z', 'below the observer'

-**P**- 'POSITION/LOCATION AT 0 / +Y / 0', 'in front of the observer'

-**K**- ‘POSITION/LOCATION AT 0 / -Y / 0’, ‘behind the observer’

-**F** ‘POSITION/LOCATION AT +X / 0 / 0’, ‘to the right of the observer’

-**V**- ‘POSITION/LOCATION AT -X / 0 / 0’, ‘to the left of the observer’

For mnemonic convenience, combinations of the above positional concepts are designated by corresponding combinations of the phonemes of the above X/Y/Z roots (-**ŢY**- and -**ŢW**- are shortened to -**Y** and -**W**):

-**PY**- ‘POSITION/LOCATION AT 0 / +Y / +Z’, ‘in front of and above the observer’

-**PW**- ‘POSITION/LOCATION AT 0 / +Y / -Z’, ‘in front of and below the observer’

-**KY**- ‘POSITION/LOCATION AT 0 / -Y / +Z’, ‘behind and above the observer’

-**KW**- ‘POSITION/LOCATION AT 0 / -Y / -Z’, ‘behind and below the observer’

-**FP**- ‘POSITION/LOCATION AT +X / +Y / 0’, ‘to the right and in front of the observer’

-**FK**- ‘POSITION/LOCATION AT +X / -Y / 0’, ‘to the right and behind the observer’

-**FY**- ‘POSITION/LOCATION AT +X / 0 / +Z’, ‘to the right and above the observer’

-**FW**- ‘POSITION/LOCATION AT +X / 0 / -Z’, ‘to the right and below the observer’

-**VB**- ‘POSITION/LOCATION AT -X / +Y / 0’, ‘to the left and in front of the observer’

-**VG**- ‘POSITION/LOCATION AT -X / -Y / 0’, ‘to the left and behind the observer’

-**VY**- ‘POSITION/LOCATION AT -X / 0 / +Z’, ‘to the left and above the observer’

-**VW**- ‘POSITION/LOCATION AT -X / 0 / -Z’, ‘to the left and below the observer’

-**FPY**- ‘POSITION/LOCATION AT +X / +Y / +Z’, ‘to the right, in front, and above the observer’

-**FPW** -‘POSITION/LOCATION AT +X / +Y / -Z’, ‘to the right, in front, and below the observer’

-**FKY**- ‘POSITION/LOCATION AT +X / -Y / +Z’, ‘to the right, behind, and above the observer’

-**FKW**- ‘POSITION/LOCATION AT +X / -Y / -Z’, ‘to the right, behind, and below the observer’

-**VBY**- ‘POSITION/LOCATION AT -X / +Y / +Z’, ‘to the left, in front, and above the observer’

-**VBW**- ‘POSITION/LOCATION AT -X / +Y / -Z’, ‘to the left, in front, and below the observer’

-**VGY**- ‘POSITION/LOCATION AT -X / -Y / +Z’, ‘to the left, behind, and above the observer’

-**VGW**- ‘POSITION/LOCATION AT -X / -Y / -Z’, ‘to the left, behind, and below the observer’

-**PK**- ‘POSITION BETWEEN/AMIDST/AMONG [IN A QUASI-PLANAR CONTEXT]’, e.g., among others in a crowded room

-**DK**- ‘POSITION BETWEEN/AMONG [IN A LINEAR UNIDIMENSIONAL CONTEXT]’, e.g., between two others in a queue

-**TK**- ‘INDEFINITE POSITION AMIDST/AMONG [IN A 3-DIMENSIONAL VOLUME]’, e.g., among a sky full of balloonists

-**ḌD**- 'POSITION/STATE INTERTWINED/INTERMINGLED/INTERMIXED IN 2-DIMENSIONAL PLANAR CONTEXT [INDIVIDUAL COMPONENTS SEPARABLE/EXTRACTABLE]', e.g., the red marbles amidst a tabletop covered with different colored marbles

-**ḌB**- 'POSITION/STATE INTERTWINED/INTERMINGLED/INTERMIXED IN 3-DIMENSIONAL VOLUME [INDIVIDUAL COMPONENTS SEPARABLE/EXTRACTABLE]', e.g., the red marbles within a jar full of different colored marbles

-**ḌV**- 'POSITION/STATE INTERTWINED/INTERMINGLED/INTERMIXED IN 2-DIMENSIONAL PLANAR CONTEXT [INDIVIDUAL COMPONENTS INSEPARABLE/PERMANENTLY COMBINED]', e.g., yellow paint spread onto a blue canvas to make a green area

-**ḌG**- 'POSITION/STATE INTERTWINED/INTERMINGLED/INTERMIXED IN 3-DIMENSIONAL VOLUME [INDIVIDUAL COMPONENTS INSEPARABLE/PERMANENTLY COMBINED]', e.g., sugar granules poured into a cup of coffee

-**NH**- **'DEGREE OF PROXIMITY/DISTANCE'** Affix: **PXM**		
BSC	**STEM 1:** (to be/manifest) a degree of linear proximity/distance in space and/or time between entities — [both the spatio-temporal expanse and the effect/consequences thereof]	**STEM 2:** (to be/manifest a) a degree of linear spatial proximity/distance/spatial interval between entities **STEM 3:** (to be/manifest a) a degree of temporal proximity/distance (i.e., elapsed time between entities)
CTE	(to be/manifest) the effect/consequences of the spatio-temporal distance/proximity between entities	
CSV	(to be/manifest) a degree of linear proximity/distance in space and/or time between entities)	
OBJ	(to be) an entity/party at a certain distance/proximity to another	

3.5 POSITIONALLY-DEFINED COMPONENTIAL PARTS OF A WHOLE

As seen above, Stem Zero of -**Ṭ**- type spatial position roots convey the meaning of "the area/vicinity around X", so that Stem Zero of the root -**K**- above would mean "the area behind or in back (of)". Nevertheless, as in Ithkuil, separate roots exist to describe positionally-defined componential parts of an entity, to distinguish "the area behind or in back of" from "the rear/back part/side of" where the latter refers to a part of the entity itself. This allows one to distinguish phrases such as "behind the box" from "the back/rear side of the box."

-**ṬF**- **'POSITIONALLY-DEFINED COMPONENTIAL PART'** Associated affix: -**S08**-		
BSC	**STEM 1:** positionally-defined part/section of an entity (e.g., side, front, top, bottom, rear, etc.); to be a positionally-defined part/section of an entity	**STEM 2:** Inalienable, inherent, inseparable, "built-in" component part/section in relation to the whole **STEM 3:** Alienable, separable, detachable component part/section in relation to the whole
CTE	The state/act of physically being/doing what a positionally-defined part/section of an entity is/does; to physically bed/do what a positionally-defined part/section of an entity is/does	
CSV	A/the (relative) spatial position/location which defines/delineates a part/section of an entity; to be a/the spatial position/location which defines/delineates a part/section of an entity	
OBJ	The entity of which the componential part/section is a part; to be the entity of which the componential part/section is a part	

Roots (and suffixes) following the same model as -**ŢF**- above are as follows:

-**KF**- 'REAR/BACK PART OF AN ENTITY [RELATIVE TO ITS DIRECTIONAL ORIENTATION OR PRIMARY EXTERNAL INTERFACE]'

-**DY**- 'TOP/PEAK/SUMMIT OF AN ENTITY [RELATIVE TO ITS TYPICAL ORIENTATION UNDER GRAVITY]'

-**DW**- 'BOTTOM/FOUNDATION OF AN ENTITY [RELATIVE TO ITS TYPICAL ORIENTATION UNDER GRAVITY]'

-**ḌY**- 'UPPER PART/"HALF" OF AN ENTITY [RELATIVE TO ITS TYPICAL ORIENTATION UNDER GRAVITY]'

-**ḌW**- 'LOWER PART/"HALF" OF AN ENTITY [RELATIVE TO ITS TYPICAL ORIENTATION UNDER GRAVITY]'

-**XW**- 'INTERIOR/INTERNAL VOLUME/"INSIDE(S)"/"INNARDS" OF AN ENTITY'

-**CL**- 'LINEAR UNI-DIMENSIONAL MIDDLE, CENTER [AS SEEN PARALLEL TO LONG AXIS OF ENTITY (OR HEIGHT-AXIS OF A "TALL" ENTITY)]'

-**CR**- 'LINEAR UNI-DIMENSIONAL MIDDLE, CENTER [AS SEEN PERPENDICULAR TO LONG AXIS OF ENTITY (OR HEIGHT-AXIS OF A "TALL" ENTITY)]'

-**ẔL**- 'PLANAR 2-DIMENSIONAL MIDDLE, CENTER [RUNNING VERTICALLY AND LATERALLY PARALLEL TO LONG (OR TALL) SIDE OF ENTITY]'

-**ẔR**- 'PLANAR 2-DIMENSIONAL MIDDLE, CENTER [RUNNING VERTICALLY PARALLEL BUT LATERALLY PERPENDICULAR TO LONG (OR TALL) SIDE OF ENTITY]'

-**ẔW**- 'PLANAR 2-DIMENSIONAL MIDDLE, CENTER [RUNNING VERTICALLY AND LATERALLY PERPENDICULAR TO LONG (OR TALL) SIDE OF ENTITY]'

-**CW**- 'CENTER POINT OF AN ENTITY, i.e., the zero-dimensional point most equidistant from all points on the periphery or surface of an entity

-**FḌ**- 'SIDE/FLANK OF AN ENTITY'

-**PF**- 'FRONT PART OF AN ENTITY [RELATIVE TO ITS DIRECTIONAL ORIENTATION OR PRIMARY EXTERNAL INTERFACE]'

-**XL**- 'INTERIOR SURFACE OR "WALL" / INTERNAL SURFACE OR "WALL" / THE INSIDE SURFACE OR "WALL" OF AN ENTITY'

-**XR**- 'EXTERIOR/EXTERNAL SURFACE/"OUTSIDE"/"SKIN" OF AN ENTITY'

The following roots are also modeled on the same root -**ŢF**-, even though they do not specify a particular spatial position relative to the whole:

-**XḌ**- 'EXTERNAL POINT-LIKE OUTWARD-FACING VERTEX/"CORNER" OF AN ENTITY'

-**XT**- 'EXTERNAL QUASI-LINEAR EDGE OF AN ENTITY'

-**XK**- 'EXTERNAL QUASI-LINEAR JOINING OF SURFACES/"SEAM" OF AN ENTITY'

-**XF**- 'EXTERNAL ATTACHMENT POINT OR FASTENING MECHANISM OF AN ENTITY', i.e., handle, hook, latch, velcro patch, knob, peg, etc.

-**ČḌ**- 'INTERNAL POINT-LIKE INTERIOR VERTEX/"CORNER" OF AN ENTITY'

-**XḌR**- 'EXTERNAL LINEAR OUTWARD-FACING "CORNER" EDGE OR "CORNER" SPACE OF AN ENTITY'

-**ČḌR**- 'INTERNAL LINEAR INTERIOR "CORNER" EDGE OR "CORNER" SPACE OF AN ENTITY'

3.6 ADDITIONAL ROOTS RELATED TO SPATIO-TEMPORAL MOTION

-ÇP- 'ACTION IN SITU / MOVEMENT IN SITU' Associated Affix: **BDM**		
BSC	**STEM 1:** (to be/manifest) an action/movement/motion in-situ (i.e., non-translative) — [both the movement itself and the entity manifesting such]	**STEM 2:** (to be/manifest) a movement of a part of a larger whole; move (in-situ) a part of an entity **STEM 3:** (to be an) cyclic/oscillating/back-and-forth/up-and-down in-situ movement
CTE	(to be) a process of in-situ motion/movement; to move in-situ	
CSV	(to be) an in-situ motion/movement itself, i.e., how the movement manifests itself; what the movement(s) consists of	
OBJ	(to be) something moving in-situ	

NOTE: for the meaning of the old FORMAL stems previously associated with this root, use the following new root:

-RÇP- 'BODILY POSE / IN-SITU BODILY MOVEMENT / DANCING'		
BSC	**STEM 1:** (to be/manifest) a bodily pose; to pose one's body and/or limbs	**STEM 2:** (to be) a state/act of in-situ (i.e., non-translative) movement of one's body and/or limbs **STEM 3:** (to be) an act of dancing
CTE	(to be) a state of being in/maintaining a bodily pose	
CSV	(to be) an act of assuming a bodily pose; to assume a bodily pose	
OBJ	(to be) the pose one assumes	

-ÇC- 'TRACK/TRAIL/WAKE' Associated Affix: **WAK**		
BSC	**STEM 1:** (to be) track/trail of passage left behind by passage of entity [both track/trail + the firmament it is in]	**STEM 2:** (to be) 1-D, 2-D or 3-D wake or front left behind by passage of entity **STEM 3:** (to be) trail, wake, or spread left behind as area of demarcation, boundary, zone, etc.
CTE	(to be) the track/trail of passage left behind by passage of an entity [focus on track/trail itself]	
CSV	(to be) the firmament/ground/substrate in which a track/trail has been left (and of which it is likely formed)	
OBJ	(to be) the entity which leaves a track/vestige/trace/trail of its passage	

-ḌÇ- 'OTHER BODILY MOVEMENT'		
BSC	**STEM 1:** (to be) an act of jumping/leaping/springing forth; to jump/leap/spring	**STEM 2:** (to be) a state/act of stretching one's muscles/limbs **STEM 3:** (to be) an act of shaking (a part of) one's body (e.g., as when shaking one's wet hair dry or an animal shaking water from its fur)
CTE	(to be) a state of being mid-leap or mid-jump or having jumped/leaped	
CSV	(to be) a physical act of jumping/leaping; to spring off one's feet/paws/legs into mid-air	
OBJ	(to be) where one jumps/leaps to, where one lands after jumping/leaping	

-ŇBY- 'FIXED PLACEMENT OR POSITIONING / PARKING / STATION-KEEPING'		
BSC	**STEM 1:** (to be) an act/instance/process of fixing in place; to fix (in place) [i.e., position something into a (quasi-)permanent orientation or placement relative to some surrounding subtrate or medium]	**STEM 2:** (to be) a state/act/ process of acitvation or change of state by placing something in a certain position; to activate a change of state by placing something in a particular position (e.g., via insertion, affixing, attaching, turning, shifting, sliding, etc.) **STEM 3:** (to be) a state/act of parking or station-keeping; to park [i.e., to place in a station-keeping or temporary storage position]
CTE	(to be) a state of being fixed in a certain position/orientation/place	
CSV	(to be) a physical act of fixing in place; to fix/position in place, to place into a fixed position	
OBJ	(to be) the position/orientation into which something is fixed	

-ḌẔ- 'STAY (PUT) / REMAIN / RETAIN / BE LEFT AS-IS'		
BSC	**STEM 1:** (to be) an act/instance/process of staying (put)/remaining in one's current spatio-temporal location/position/ state/situation for the short-term context-at-hand; to stay (put), to not change one's current location/position/state/situation for the short-term context-at-hand; to stay/remain (for the short-term)	**STEM 2:** (to be) a state/act/ process of staying/remaining in/retaining one's current spatio-temporal location/position/ state/situation for the long-term (i.e., beyond the original temporal context); to stay/remain/retain (in the long term) **STEM 3:** (to be) a state/act of being left in a particular spatio-temporal location/position/ state/situation with no further involvement by the initial party which caused the entity to be situated thus (and without whose help or the help of another party, the entity so left cannot change their status); to leave/be left (behind) as is
CTE	(to be) a state of not having changed one's spatio-temporal location/position/ state/situation during the short-term context-at-hand	
CSV	(to be) a physical act of staying/remaining in one's spatio-temporal location/position/ state/situation for the short-term context-at-hand	
OBJ	(to be) the spatio-temporal location/position/ state/situation in which one stays (put)/remains for the short-term context-at-hand	

3.7 TEMPORAL CONCEPTS

-LN- 'TIME OF DAY'			
	STEM 1	**STEM 2**	**STEM 3***
BSC	(to be) daytime (= portion of the 24-day when the sun is above the horizon)	(to be) evening or nighttime (= portion of the 24-hour day when the sun is below the horizon)	(to be) a particular "o'clock"-time of the day as named by the hour [use numerical roots or affixes to specify the number of hours since midnight] (includes both the time and the event occurring then)
CTE	(to be) the elapsing/passing of time during the daytime; to spend/pass the daylight hours	(to be) the elapsing/passing of time during the evening/nighttime; to spend/pass the evening/nighttime hours	(to be) the measuring/delineation of time as specified by a particular "o'clock"-time of day
CSV	(to be) the state of being in daylight; to occur/manifest during daylight hours	(to be) the state of being nighttime; to occur/manifest during evening/nighttime hours, to be/occur at night	(to be) the state of being/occurring at a particular "o'clock"-time of day
OBJ	(to be) that which occurs/passes during daylight hours	(to be) that which occurs/passes during evening/nighttime hours	(to be) the event which occurs at or is specified by a particular "o'clock"-time of day

*for specific times, e.g., 10:15 a.m. and 34.3 seconds, use Stem 3 with an appropriate numerical affix, followed by numerical stems with COO affix plus the appropriate stems of the -**RW**- root below declined in the PARTITIVE case. Alternately, one may use numerical stems with the various degrees of the ELA affix.

-RV- **TEMPORARY DURATION** Associated Affix: **TD1**		
BSC	(to be) a particular amount of elapsed time during which something occurs/exists; for something to occur/exist lasting/enduring a particular amount of time	**STEM 1:** a moment/instant **STEM 2:** a "while" **STEM 3:** a portion of a day
CTE	(be) the event(s) occurring during a particular amount of elapsed time	
CSV	(be) the duration of elapsed time [regardless of what may occur/exist during that period]	
OBJ	(be) the "volume" of spacetime during/in which something occurs/exists	

NOTE: for the meaning of the old FORMAL stems previously associated with this root, use the new root -**RW**- shown below:

Specifications for the following six roots are modeled after the root -**RV**- 'TEMPORARY DURATION' above

-RW- **SHORT-TERM CONVENTIONALIZED TIME PERIOD**
STEM 1: (to be) a second
STEM 2: (to be) a minute
STEM 3: (to be) an hour

-Ř- CONVENTIONALIZED TIME PERIOD
STEM 1: (to be a) day (= 24-hour period)
STEM 2: (to be a) week
STEM 3: (to be a) lunar period / month

NOTE: for the meaning of the old FORMAL stems previously associated with this root, use the following new root:

-RḌ- CALENDRICAL TIME PERIOD Associated Affix: **TD2**
STEM 1: (to be a) calendrical day, designated day (for planning/scheduling purposes)
STEM 2: (to be a) calendrical week, designated week (for planning/scheduling purposes)
STEM 3: (to be a) calendrical month, designated month (for planning/scheduling purposes)

-RN- LONGER TIME PERIOD Associated Affix: **LTD**
STEM 1: (to be a) calendrical year, designated year (for planning/scheduling purposes)
STEM 2: (to be a) calendrical decade, designated decade (for planning/scheduling purposes)
STEM 3: (to be a) designated lifetime of an individual (for planning/scheduling purposes)

NOTE: for the meaning of the old INFORMAL stems previously associated with this root, use the above stems with the DGR/8 affix.

-ŘṬ- LONG-TERM TIME PERIOD Associated Affix: **LGD**
STEM 1: (to be) calendrical century
STEM 2: (to be) calendrical millennium
STEM 3: (to be) calendrical ten-thousand year period

NOTE: for the meaning of the old INFORMAL stems previously associated with this root, use the above stems with the DGR/8 affix.

-ŘŇ- **EPOCH-LENGTH TIME PERIOD** Associated Affix: **EPC**
STEM 1: (to be) a geologic age
STEM 2: (to be) a geologic epoch
STEM 3: (to be) a geologic era

NOTE: for the meaning of the old INFORMAL stems previously associated with this root, use the above stems with the DGR/8 affix.

-RP- **'CYCLE / REGULARITY / FREQUENCY / PERIODICITY / ITERATION'** Associated Affix: **FRQ**			
	STEM 1	**STEM 2**	**STEM 3**
BSC	(to be) something having a cycle or pattern of regularity; to be something cyclic, regular or periodic [both the entity and the cyclic pattern]	(to be) something with a particular cycle or pattern of regularity; to be something with a particular cycle or regular period [both the entity and the cyclic/periodic pattern]	(to be) something manifesting iteration/recurrence, i.e., happening more than once; to be something iterative/recurrent [both the entity and the recurrence]
CTE	(to be) a pattern of cyclic/regular/periodic activity [focus on pattern as distinct from other/different potential patterns)	(to be) the particular frequency/periodicity of a phenomenon (= the rate of its recurrences/manifestations); to recur; to repeat (in terms of temporal periodicity)	(to be) a pattern of iteration/recurrence [focus on the fact that iteration/recurrence demonstrates a potential pattern or precedent for further recurrence)
CSV	(to be) a process of cycling/periodicity [focus on significance that a pattern exists/occurs]	(to be) a process of cycling/recurring with a particular period [focus on the significance that such a pattern exists/occurs]	(to be) a process of iteration/recurrence [focus on the fact that there's been a recurrence]
OBJ	(to be) the object/entity which displays cyclic/regular/periodic activity	(to be) the object/entity which displays the particular cycle/recurring/periodic pattern	(to be) the event/entity which has recurred

-KT- **PAST / PRESENT / FUTURE** (based on Metaphor of a Progressive Linear Spacetime-line) Associated Affix: **TPP**		
BSC	**STEM 1:** (to be) a past event/occurrence/entity (relative to the contextual present); to be/occur in the past [both the event/entity and its place on a linear timeline relative to the contextual present]	**STEM 2:** present event, entity, occurrence **STEM 3:** future event, entity, occurrence
CTE	(to be) a past event [focus on what occurred, not when]	
CSV	(to be) when a past event occurred [focus on when it occurred, not what occurred]	
OBJ	(to be) the entity/object/circumstance impacted/affected by a past event/occurrence/entity	

NOTE: The above stems may be used in spatial contexts as well, in which case English translations might differ depending on context, e.g., 'previous', 'former', 'once-', 'one-time', 'here', '...at hand', 'there', '-to-come', 'expected/awaited', etc.

-KM- **'TIMELINESS / PUNCTUALITY / EARLINESS / LATENESS'** Associated Affix: **TME** (formerly **LAT**)		
	STEM 1	**STEM 2**
BSC	(to manifest) a degree of timeliness/punctuality; the extent to which something is timely/on-time [use a quantitative affix such as EXN, EXD, SUF, etc., to specify]	(to be) something that is early, sooner-than-expected, sooner-than-required
CTE	(to be) that which manifests a degree of timeliness/punctuality	(to be) that which is early
CSV	(to be) a state of timeliness/punctuality, being on time	(to be) a state of earliness, being sooner-than-required; to be early
OBJ	(to be) that to/with which one is (intended to be) punctual	(to be) the event/place/situation at/to which one is (or intends to be) early
STEM 3: (to be something that is) late, untimely, tardy — Specification pattern is modeled after Stem 2		

-RŠ- **'STAGE / STEP / PHASE OF A PROCESS'** Associated Affix: **STG**		
BSC	**STEM 1:** (to be a) step/stage/phase in a process [both the entity/entities/events/acts constituting the portion of the larger process and the process/state of being subdivided from the larger process	**STEM 2:** (to be) a degree/grade or point on a progressive/scalar gradient **STEM 3:** (to be a) section, sub-unit, discernible/identifiable/differentiated "stretch" or portion of a progressively/successively structured entity/phenomenon
CTE	(to be the) entity/entities/activity/activities/act(s) within (or which constitute) a step/stage/phase of a process; to constitute the particulars of a stage in a process	
CSV	(to be) the act/state/process of being/functioning as a step /stage/phase of a process; to perform a step in a process	
OBJ	(to be the) process which is divided up into steps/stages/phases	

-TPH- **'SCHEDULE / TIMETABLE / ITINERARY /CALENDAR'**		
BSC	**STEM 1:** (to be) an act/process/instance of using/making/relying upon a schedule/timetable; to use/rely on a schedule/timetable [i.e., a written/graphic listing of sequential events delineated by a time index]	**STEM 2:** (to be) an act/process/instance of using/making/relying upon an itinerary; to use/rely on an itinerary **STEM 3:** (to be a) an act/process/instance of using/making/relying upon a calendar
CTE	(to be) a state of being on a schedule, of being scheduled	
CSV	(to be) a physical act of scheduling; to put/place/input on/in a schedule, to schedule	
OBJ	(to be) a schedule	

-LPŠ- ‘REFRESH / RENEWAL / UPDATE’		
BSC	**STEM 1:** (to be) an act/process/state of needing to be refreshed/replenished; to be in need of refreshment/replenishment/resupply [i.e., to make something return to having its initial appearance/condition/supply] CPT version = refresh/replenish/resupply	**STEM 2:** (to be) an act/process/instance of renewing/reviving/ renewal/revival;; to renew/revive [i.e., to make something return to the state it had when new] **STEM 3:** (to be a) an act/process/instance of updating, bringing something up-to-date; to update [i.e., to correct aspects of an entity that have become obsolete/inaccurate]
CTE	(to be) a state of needing to be refreshed/replenished/resupplied CPT version = state of being refreshed	
CSV	(to be) a physical act of refreshing/replenishing/resupplying; to refresh/replenish/resupply	
OBJ	(to be) what is in need of refreshment/replenishment CPT version = what is/has been refeshed/replenished	

4.0 PSYCHOLOGICAL AND SOCIOLOGICAL PHENOMENA

-SL- ‘THOUGHT / CONTEMPLATION / REASON / INTELLECT’		
BSC	STEM 1: (to be) an act of thinking/cogitating; to think/cogitate	STEM 2: (to be) something one is considering/mulling/contemplating; to consider, mull, contemplate, take into account STEM 3: (to be) an act of reasoning; to reason, use one’s intellect [= employ a strict process of integrating logic and experience when thinking/analyzing]
CTE	(to be) the state/process of thinking/cogitating	
CSV	(to be) the thought one is thinking	
OBJ	(to be) the basis/trigger for a thought; that which makes one think of something	

DERIVATIONS: ponder, deliberate/deliberation, contemplate, mull, concentrate, theory, hypothesis, judge, conclude/conclusion

This root is also used as the **PROPOSITIVE** (**PPV) Bias Affix:** *‘what if…’ ‘It could be that….’ ‘Consider this: …’ ‘Posit the following: …’ ‘Assume for the sake of argument that….’*

-KSL- ‘IDEA / ANALYSIS / ABSTRACTION / CONCEPT’
Stem 1: (to be) an act of thinking up/of an idea; to brainstorm; to come up with an idea CSV = an idea
Stem 2: (to be) an act of analysis; to analyze
Stem 3: (to be) an act of abstraction/; to abstract CPT = conceptualization/conceptualize CSV = a concept